

Detecting Potential Tornado damage

By

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Introduction

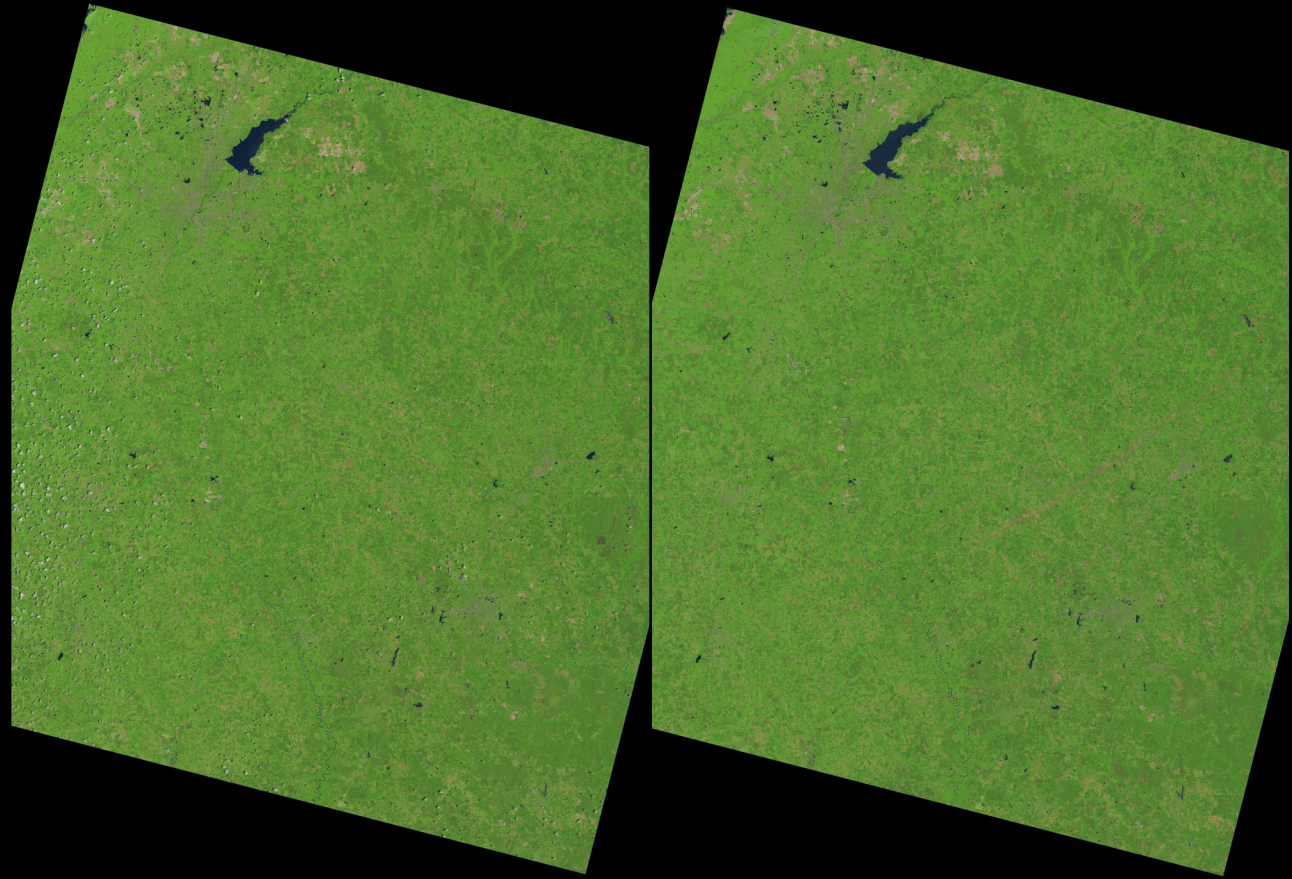
- Tornadoes are some of the most destructive and commonly occurring natural disasters in the continental United States and similarly to other natural disasters like forest fires, tornadoes will become more common as climate change continues. Specifically, it increases the duration of tornado season worldwide, resulting in an estimated increase of stronger US tornado days anywhere from 3.8% to 12.7% by the 2090s (Lee, C.C. 2011). This highlights the need to understand how destructive tornadoes can be as they will occur more often in the future.
- Given that most of the resources, ENVI and ArcGIS excluded, used in this project are freely accessed. It is incredibly useful and cost effective to use Satellite imagery to gauge the impacts of natural disasters in communities around the world, help with aid estimation and authenticate insurance claims.

Objectives

- Detect potentially damaged areas affected by tornadoes based on change in vegetation.
- Compare damaged areas to on the ground records of damage.
- Expand upon the utility of remotely sensed techniques and Ground truth data gathering.

Study area

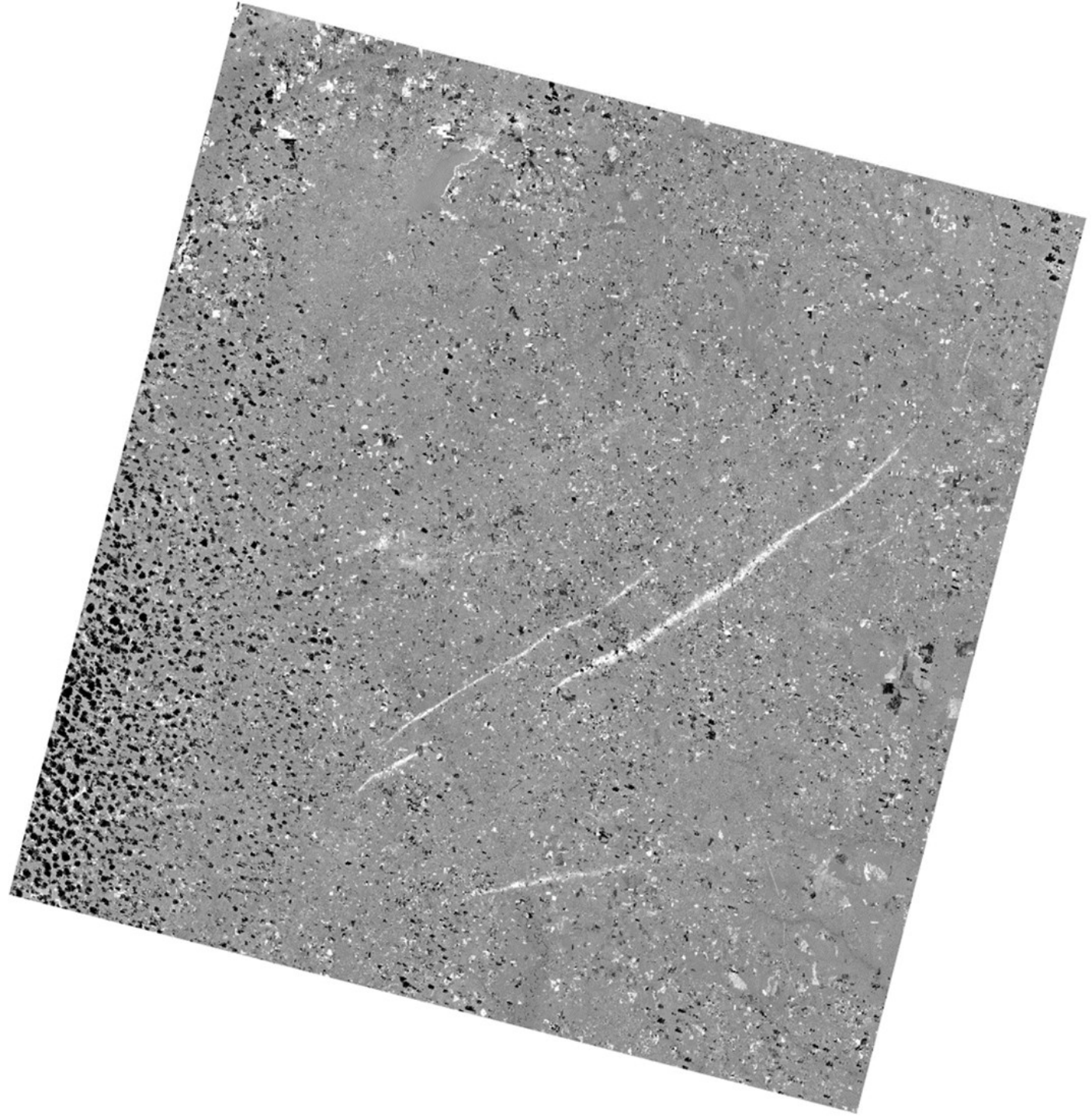
- **Hope-Sartinville–
Bassfield, Mississippi
area**
- Left May: 25th, 2019
- Right: May 11th, 2020



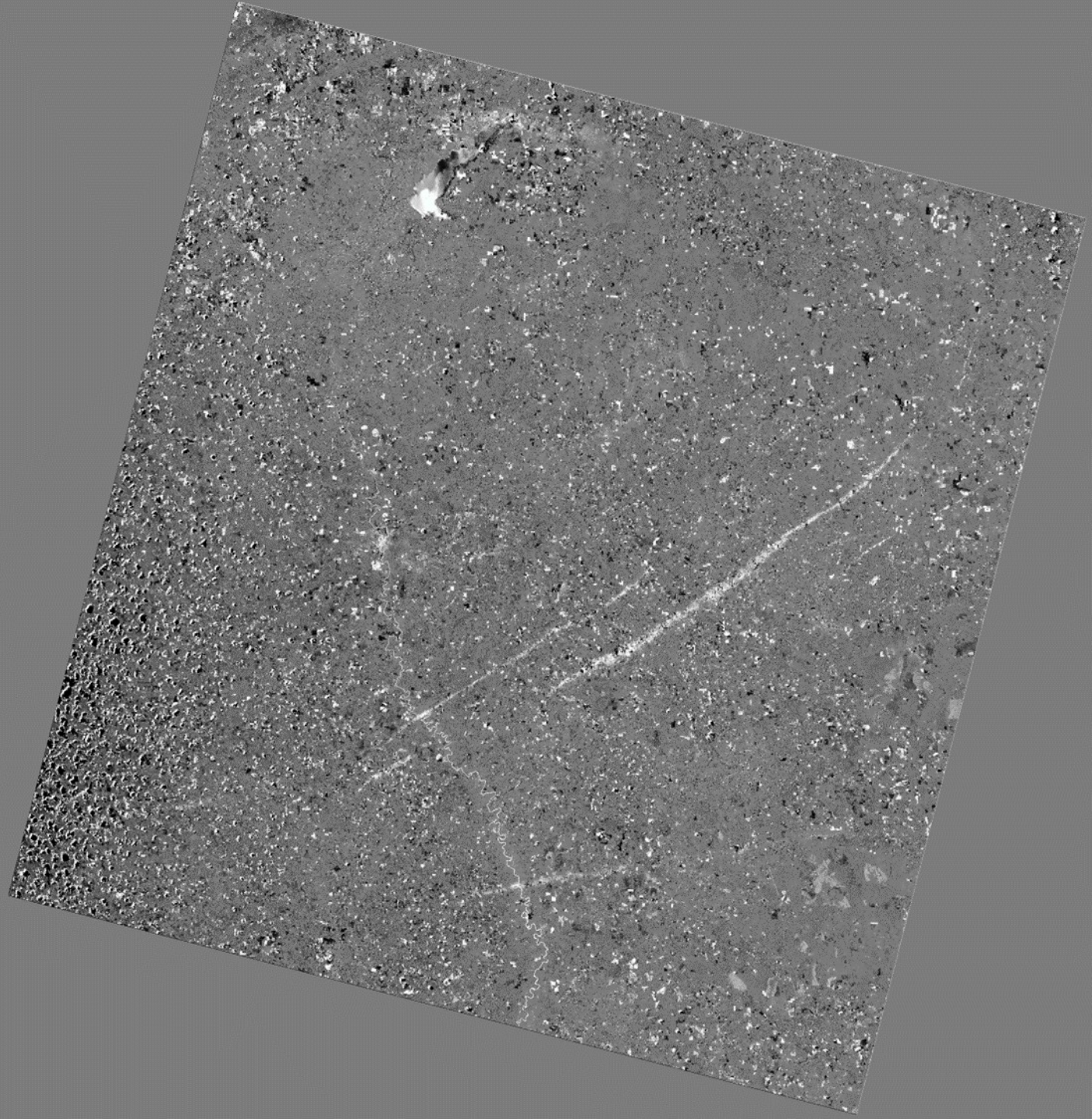
Methods

- Normalized Difference Vegetation Index (NDVI)
- Tasseled Cap Greenness
- Pixel classification based on change thresholds (b1 – b2)
- Heads up classification of threshold pixels
- NOAA Data comparison of tiger files to find trends in NOAA data

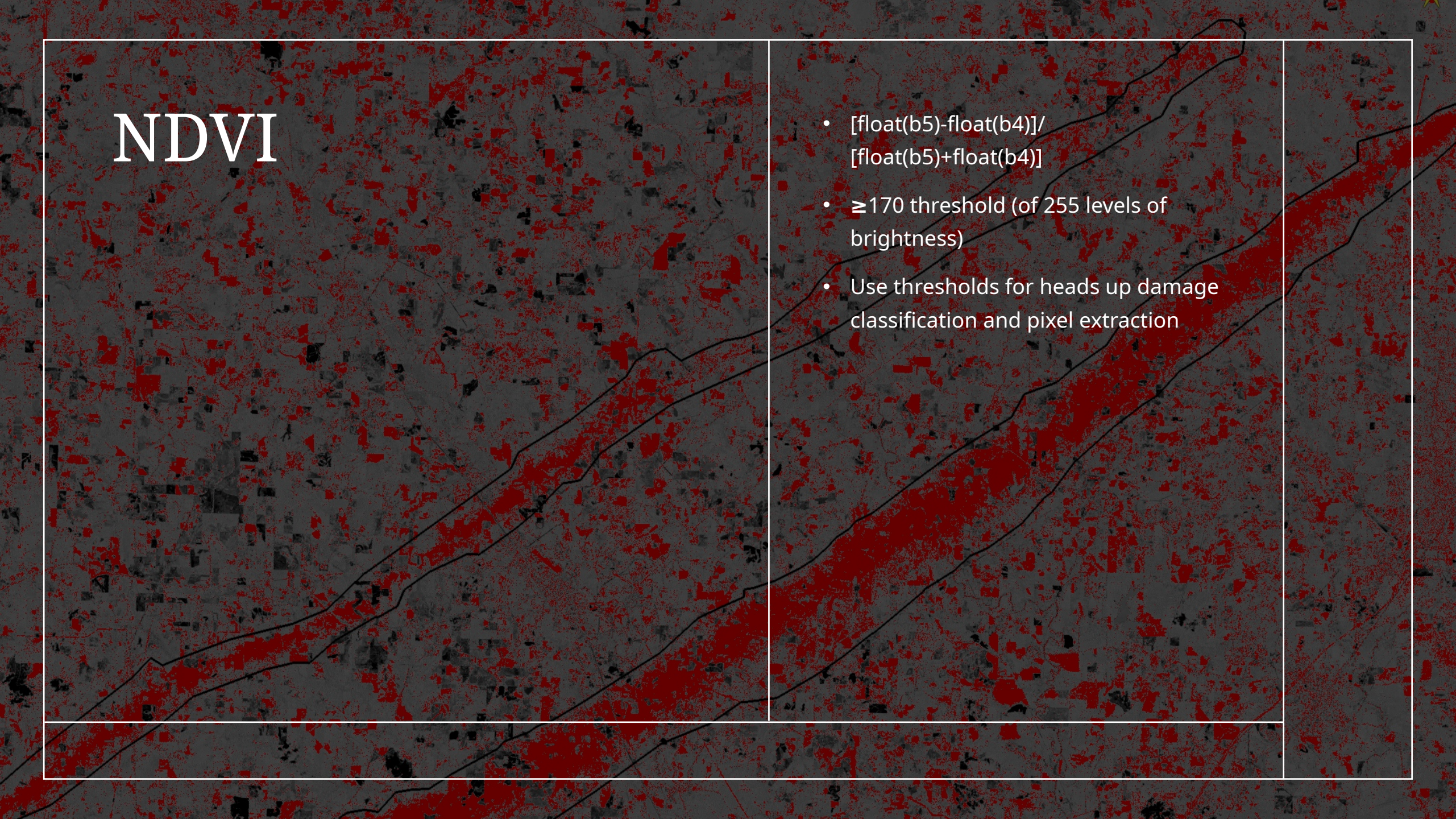
NDVI Change



Tasseled Cap Change



NDVI



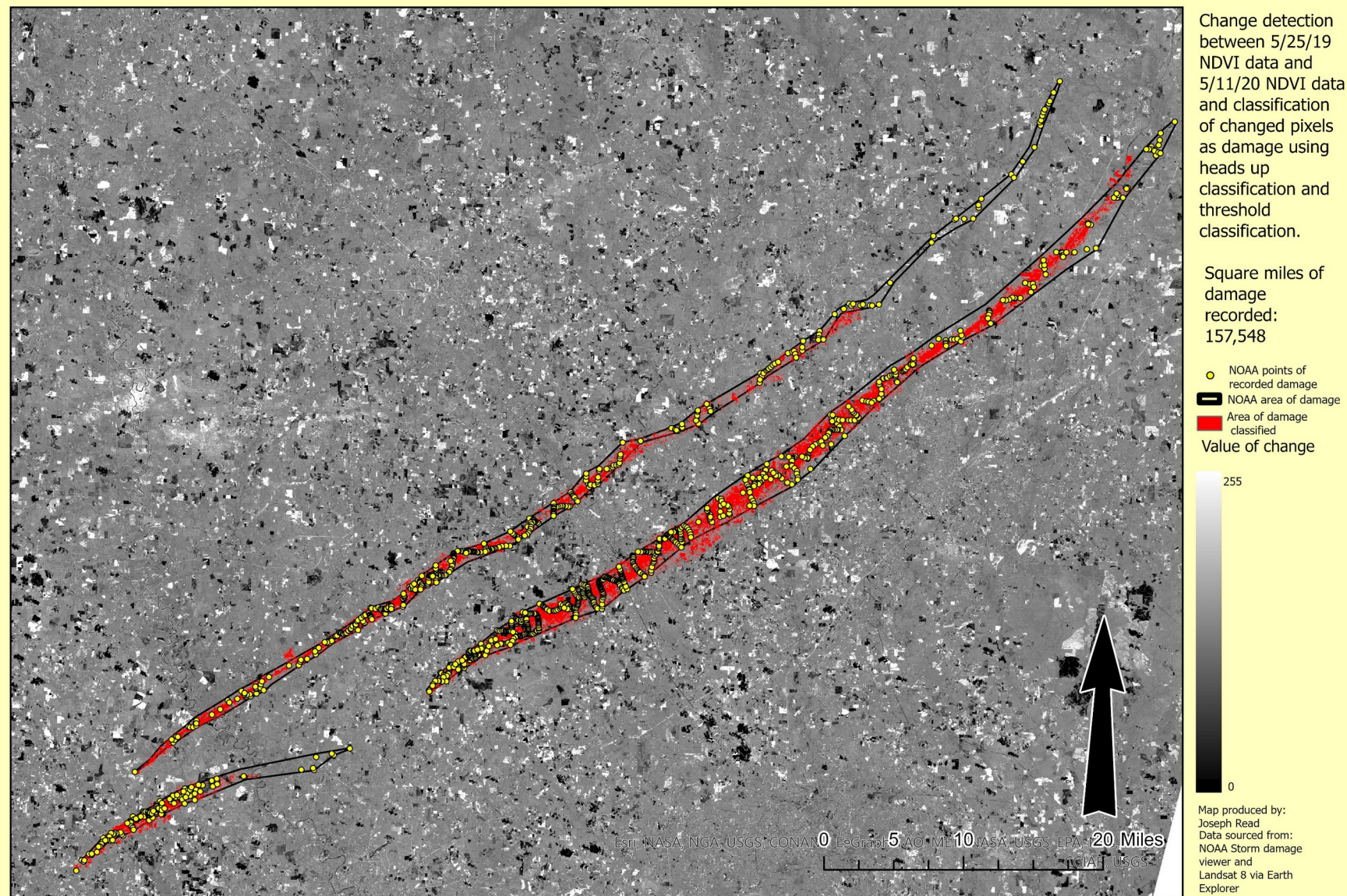
- $\frac{[\text{float}(\text{b5}) - \text{float}(\text{b4})]}{[\text{float}(\text{b5}) + \text{float}(\text{b4})]}$
- ≥ 170 threshold (of 255 levels of brightness)
- Use thresholds for heads up damage classification and pixel extraction

Tassel Cap

- Greenness: $b1*(-0.2941)+b2*(-0.243)+b3*(-0.5424)+b4*0.7276+b5*0.0713+b6*(-0.1608)$
- ≥ 140 threshold (of 255 levels of change)
- Use threshold for heads up damage classification and extraction

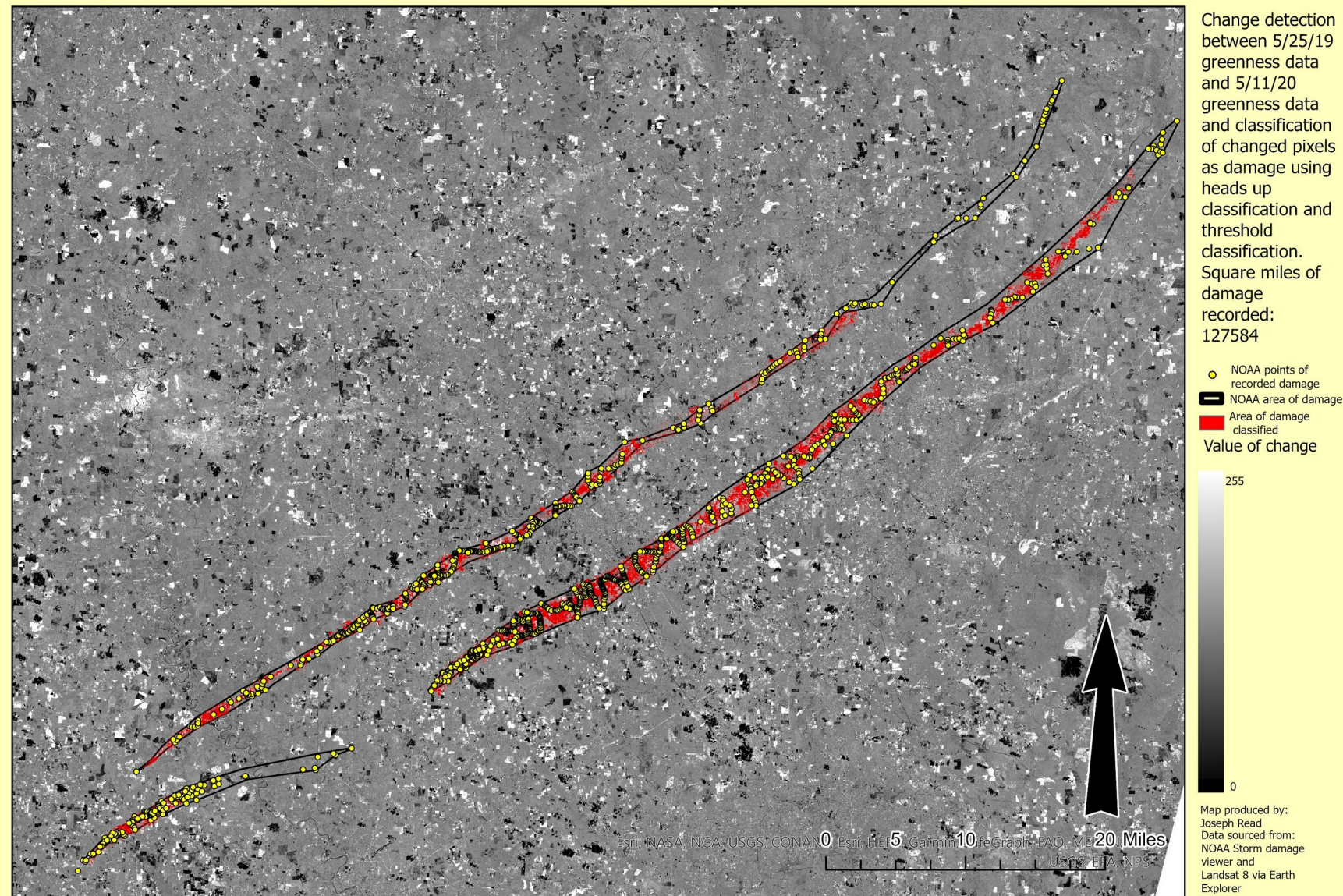
Results: NDVI

2020 Hope Tornado damage detection using ENVI and ArcGIS Pro: NDVI



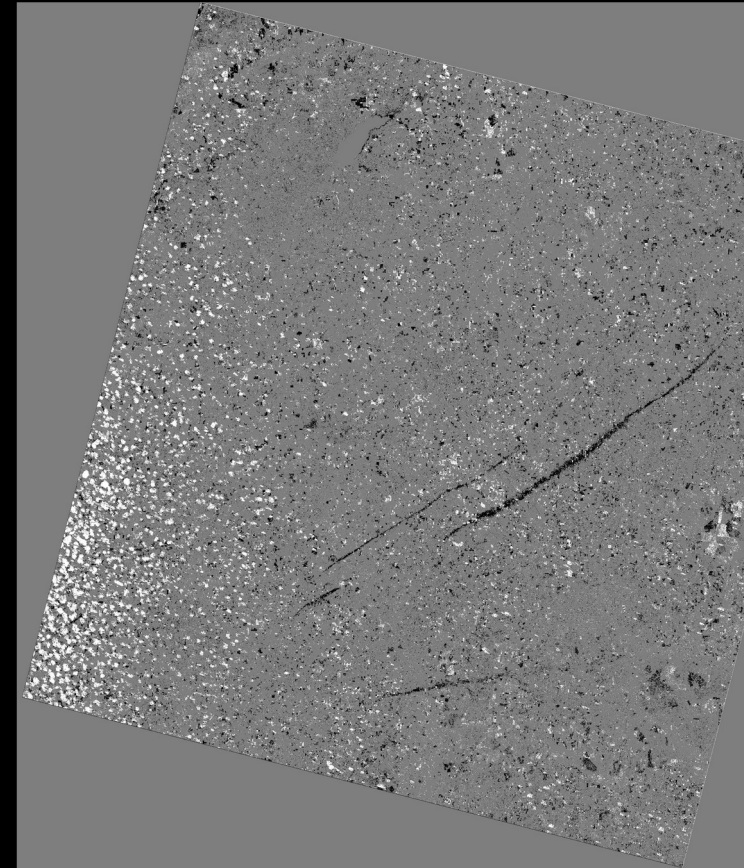
Results: Tassel Cap

2020 Hope Tornado damage detection using ENVI and ArcGIS Pro: Tassel Cap



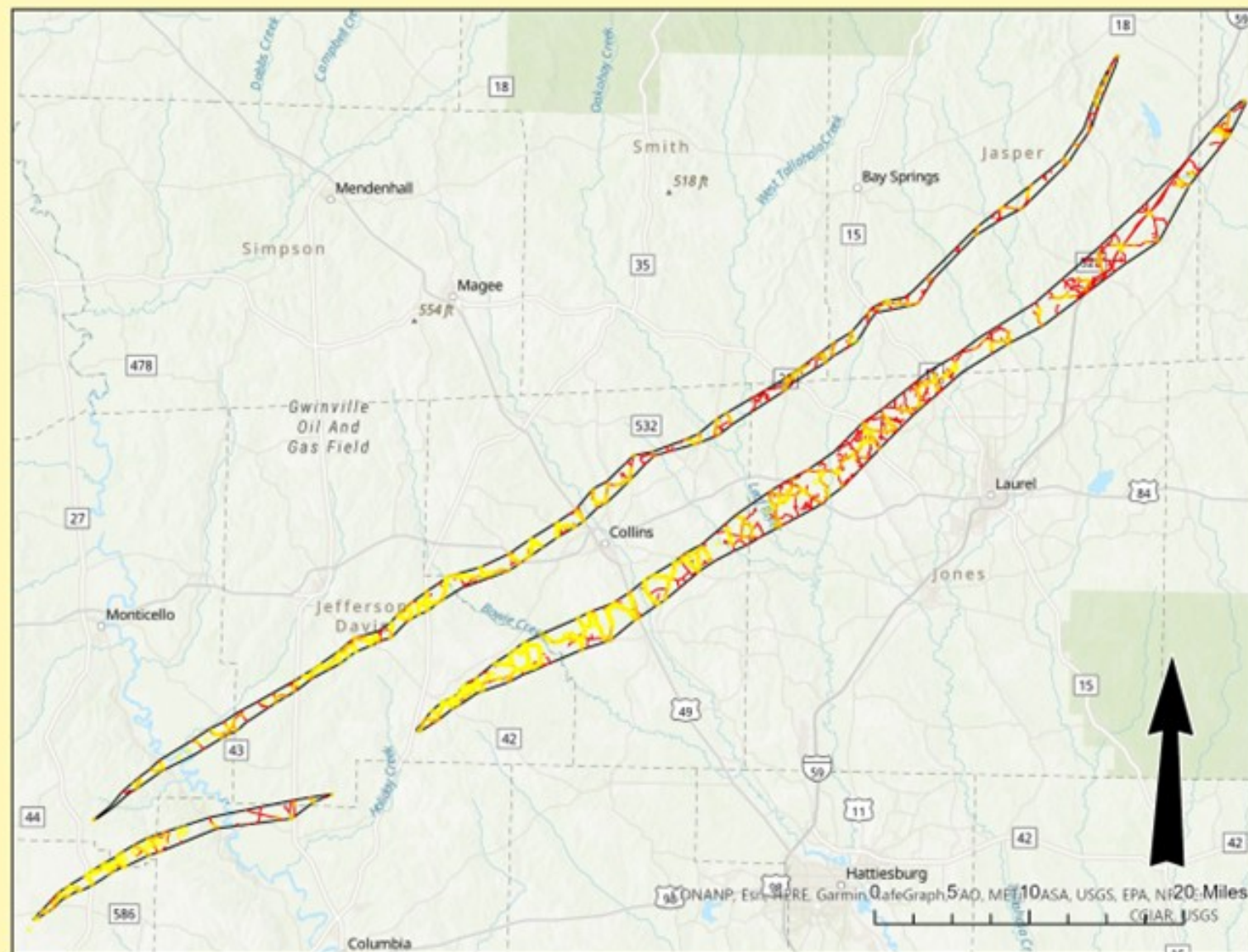
Discussion

- NDVI has a more dynamic range in change and encompassed more pixels of tornado damage. 157,548 square miles was detected as damaged using this method.
- The Tassel Cap analysis was less encompassing. 127,584 miles was detected as damaged using this method.
- PCA even less encompassing.



Data comparison Biases

2020 Hope Tornado damage recorded by NOAA along roads



Conclusions

A black and white photograph of a satellite in orbit. The satellite is a rectangular structure with several large, rectangular solar panels extended from its main body. It is positioned against the backdrop of the Earth's curved horizon and the dark void of space. The satellite's orientation is such that it appears to be moving from left to right across the frame.

- Changes to land from tornadoes can be observed from space and has the potential to be more encompassing on recording damage cheaper and more quickly than in person analysis.
- Some information such as knowing which fields have gone fallow is essential in knowing what change is purely from tornadoes given there is a year difference in between capture dates.
- Satellite images are good for predicting areas damaged by tornadoes.
- NDVI is better at finding slight change in plant health than tassel cap greenness in the methods described above.

Sources

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